Reg. No. | | | | | | | | | | | [18AS72]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

COMPUTATIONAL FLUID DYNAMICS AND AERODYNAMICS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1. a. Explain the integral and differential forms of control volumes used in deriving governing equations of fluid mechanics. **06**

b. Derive the continuity equation for a differential control volume moving with the flow. **04**

c. What is Reynolds averaging used in turbulence modeling? **04**

2. a. Explain the physical importance of **04+04**
- **i.** Elliptic partial differential Equations.
- **ii.** Hyperbolic linear partial differential equation.

b. Classify the following equations as elliptic , parabolic or hyperbolic **03+03**
- **i.** $\frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} = 0$
- **ii.** $\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial t^2}$

3. a. Find the finite difference discretization for the following equations **06**

$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = 0$
- **i.** using forward difference for time and central difference for space.
- **ii.** using forward difference for time and backward difference for space.

b. Explain the advantages and disadvantages of structured and unstructured grids. **04**

c. Explain the cell centered and cell vertex finite volume. **04**

OR

4. a. Explain the implementation of finite volume method for a scalar conservation law. **14**

5. a. Explain the advantages and disadvantages of following **04**
- **i.** Implicit schemes
- **ii.** Explicit schemes

b. Explain the time stepping schemes **10**
- **i.** Forward Euler scheme
- **ii.** Backward Euler scheme

OR

6.	**a.**	What are applications of Runge Kutta schemes?, Explain the derivation of a second order Runge-Kutta scheme.	**10**
	b.	Explain the flow condition suitable for the application of implicit or explicit scheme.	**04**
7.	**a.**	Explain what is the purpose of a staggered grid? Draw example for staggered grid control volume.	**05**
	b.	Explain the difference between compressible and incompressible flow solvers.	**05**
	c.	Explain the working of Riemann solvers used in compressible flows.	**04**

OR

8.	**a.**	Describe the algorithm for SIMPLE Method with a flow chart for solving the system of equations.	**10**
	b.	Why is numerically solving incompressible flows difficult compared to compressible flows ?	**04**